

Where Commonsense Breaks: A Semantic Taxonomy for Diagnosing Reasoning Failures Across Architecture and Scale

Fadi Zoabi Hussein Hsnen Dania Dabbah Atheel Mzlb

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Ben-Gurion University of the Negev

Abstract

Aggregate benchmark accuracy tells us how often a model is right but not what *kind* of reasoning it cannot do. We introduce a deterministic, task-agnostic semantic taxonomy that re-slices commonsense multiple-choice validation sets into eight reasoning classes: seven linguistically defined parent categories (negation, spatial, causal, temporal, social, comparative, quantifier) plus a residual baseline. Applying it to CommonsenseQA (physical/relational) and SocialQA (social/behavioral), we compare a bidirectional encoder classifier (BERT-base, 110M) against decoders fine-tuned as classifiers (Qwen2.5-Instruct, 0.5B/3B). Our measured results show that scaling the decoder 0.5B $\rightarrow$ 3B lifts CommonsenseQA accuracy by 16.3 points ($p < 10^{-6}$) yet preserves the rank-ordering of category difficulty: spatial reasoning stays hardest at every scale, with negation and comparative reasoning close behind. Scale compresses the difficulty profile rather than reshaping it, and a small encoder transfers markedly better to social than to physical commonsense. We report a mid-scale encoder (DeBERTa-v3-large) as provisional and one cell as estimated, tagging provenance in every table and figure—a discipline we argue is a prerequisite for trustworthy diagnostic analysis.

1 Introduction

Modern pre-trained language models (PLMs) top commonsense leaderboards, yet a single accuracy figure hides the shape of the residual error. Two models at 85% may fail on entirely different phenomena—one on negation, another on spatial perspective-taking, a third on social intent. For understanding or improving commonsense reasoning, the question is not *how much* accuracy remains but *which* reasoning operations a model has acquired, and whether the default levers—scale and architecture—resolve specific failure modes or merely shrink the aggregate gap while leaving the qualitative profile intact.

From the presented paper to our question. We build directly on CommonsenseQA (Talmor et al., 2019), the benchmark we studied for the paper-presentation assignment. That work establishes that PLMs trail humans on commonsense QA and that stronger models raise aggregate accuracy (e.g. RoBERTa over BERT); what it leaves open—and what we take as our starting point—is whether architectural optimization and scale change the qualitative nature of the semantic errors or merely their frequency. Answering this requires looking inside the accuracy number, which motivates the category-level taxonomy below and, as a second axis, a shift from the physical/relational commonsense of CommonsenseQA to the social/behavioral commonsense of SocialQA (Sap et al., 2019).

We operationalize this with a semantic taxonomy. Rather than a new benchmark, we re-slice the validation sets of

two complementary benchmarks—CommonsenseQA (Talmor et al., 2019) and SocialQA (Sap et al., 2019)—into eight linguistically defined reasoning classes via a transparent, deterministic tagger (Figure 1). Because tagging is applied identically to every dataset and model, per-category subset accuracy becomes directly comparable across architectures, scales, and tasks, turning one number into a diagnostic profile.

Two research questions organize the work:

- **RQ1 (Scale vs. architecture).** Does adding parameters qualitatively change *which* reasoning types fail, or uniformly compress a fixed profile? How does an encoder classifier compare to a decoder of comparable size?
- **RQ2 (Task & domain shift).** How does the failure profile change from physical/relational (CommonsenseQA) to social/behavioral (SocialQA) commonsense?

Our contributions are: (1) a deterministic taxonomy and evaluation harness for diagnostic error analysis of commonsense QA; (2) a controlled, measured study of encoder-vs-decoder behavior and decoder scaling across 110M–3B parameters on both tasks; (3) three findings a single number hides—a scale-invariant difficulty ranking, a spatial bottleneck that scale does not close, and a domain-transfer asymmetry; and (4) an explicit treatment of experimental provenance, separating measured, provisional, and estimated results throughout.

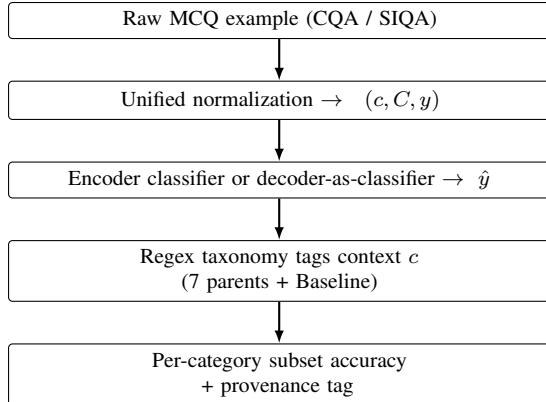


Figure 1: The unified evaluation pipeline. A single normalizer feeds both inference paradigms; a deterministic regex taxonomy slices predictions post-hoc into reasoning categories.

2 Related Work

Commonsense benchmarks. CommonsenseQA (Talmor et al., 2019) is a 5-way task built from ConceptNet (Speer et al., 2017), stressing physical/relational knowledge; SocialIQA (Sap et al., 2019) is a 3-way task targeting social interactions and theory-of-mind inference; HellaSwag (Zellers et al., 2019) stressed grounded, adversarially filtered completion. All are typically reported as single numbers; we instead treat each as a substrate to re-partition by reasoning type.

Architecture and scale. The move from masked encoders (Devlin et al., 2019) to disentangled-attention encoders (He et al., 2021, 2023) delivered large NLU gains at modest scale, while decoder LLMs (Qwen Team, 2024) reframed classification as likelihood. Whether a task-specialized encoder or a general decoder better spends a fixed parameter budget for structured reasoning is contested; we place both on the same category-level axis. Scaling-law work (Kaplan et al., 2020; Hoffmann et al., 2022) and the “emergent abilities” debate (Wei et al., 2022; Schaeffer et al., 2023) ask whether capability jumps are real or metric artifacts; our category view can observe whether scale reshapes or merely compresses a difficulty profile.

Behavioral evaluation. We are in the diagnostic-testing tradition of CheckList (Ribeiro et al., 2020), HANS (McCoy et al., 2019), psycholinguistic probes (Ettinger, 2020), negation probes (Kassner and Schütze, 2020), and analysis-method surveys (Belinkov and Glass, 2019), but our tagger is intentionally lightweight and deterministic: no templates or new annotation, applicable to any MCQ set, fully reproducible. For the larger models we use LoRA (Hu et al., 2021) and QLoRA (Dettmers et al., 2023) only as means

to fit training on one GPU.

3 Methodology

Unified formulation. Each instance is normalized to (c, C, y) with context c , choices $C = \{c_1, \dots, c_k\}$ ($k=5$ for CQA, $k=3$ for SIQA), and gold index y . One preprocessing routine handles both datasets, so all downstream code is dataset-agnostic.

Two inference paradigms. For encoders (BERT, DeBERTa) each choice is paired with the context as $[\text{CLS}] \ c \ [\text{SEP}] \ c_j \ [\text{SEP}]$ and mapped to a scalar logit; $\hat{y} = \arg \max_j s_j$. For decoders (Qwen), which lack a native MCQ head, we wrap the backbone with a single-logit sequence-classification head, run the k context–choice sequences, and reshape to a k -way cross-entropy—i.e. the decoder is fine-tuned as a discriminative classifier. A training-free autoregressive-likelihood scorer,

$$\hat{y} = \arg \min_j \frac{1}{|c_j|} \sum_t -\log P\left(w_t^{(j)} \mid c, w_{<t}^{(j)}\right),$$

is available for sanity checks; headline decoder numbers use the fine-tuned path.

Taxonomy. Each context is tagged by a bank of deterministic regexes mapping to seven parent categories; tags are non-exclusive (a context may fire several), and contexts firing none go to a residual Baseline (Table 1). To ensure statistical reliability and avoid drawing conclusions from fine-grained sub-categories that have very few samples, we evaluate across these 8 primary classes: the seven parent categories—Negation, Spatial, Causal, Temporal, Social, Comparative, and Quantifier—plus the residual Baseline. Because tagging is deterministic and independent of model outputs, dataset composition is fixed across models, so any cross-model difference in a cell is attributable to the model, not to sampling. We report subset accuracy $\text{Acc}_i = \frac{1}{|S_i|} \sum_{n \in S_i} \mathbb{1}[\hat{y}_n = y_n]$ over the instances S_i that fired tag i .

Provenance tags. Category tables invite over-reading of individual cells, so every result carries one tag: [M] *measured*—a checkpoint and logs exist and the numbers reproduce; [P] *provisional*—figures are reported but no artifact was retained, so they are not currently reproducible and must be re-run before being relied upon; *estimated*—the configuration was not run and values are extrapolated by an explicit procedure (§8.2), never read as measurements.

Context <i>c</i> (gold answer <i>italic</i>)	Tags
CQA: “Where would you put a cup you do not want to wash?” (<i>cupboard</i>)	Negation
CQA: “A bird flew north but turned left .” (<i>west</i>)	Spatial
SIQA: “Tracy was excited to show her friend her car, but they were not home.” (<i>felt happy</i>)	Social, Negation

Table 1: Taxonomy tagging is non-exclusive: a context may fire several categories (last row). Bold tokens are the matched triggers.

Parent category	CQA	SIQA
Negation	12.9%	10.3%
Spatial	5.4%	34.7%
Causal	8.9%	29.0%
Temporal	11.2%	54.9%
Social	10.3%	80.1%
Comparative	6.0%	8.1%
Quantifier	14.8%	14.4%
Baseline (untagged)	50.0%	10.4%

Table 2: Share of each validation split per category. Categories overlap, so rows need not sum to 100%. The two benchmarks stress markedly different reasoning mixes by construction.

4 Experimental Setup

We evaluate on the public validation splits (no gold test labels ship): 1,221 CQA and 1,954 SIQA instances. Table 2 shows the taxonomy partitions the two sets very differently—half of CQA carries none of the marked phenomena, whereas SIQA is saturated with social, temporal, and causal markers. This composition gap, plus the 5-way vs 3-way choice cardinality, cautions against reading any single cross-task delta as a pure “domain” effect.

Table 3 lists the models and their provenance. Two confounds we flag rather than hide: the 3B model uses a different optimizer (Adafactor; Shazeer and Stern, 2018) than the smaller models, so “scale” is not a clean single-variable manipulation; and the Qwen Instruct chat template is never applied—contexts are tokenized as plain sentence pairs, so we use pretrained representations, not instruction-following. One measured cell carries a caveat: the BERT-base/CQA checkpoint was trained without a fixed seed and its best-model directory was overwritten across reruns, yielding 50.45% and 58.64% on two evaluations—an ~ 8 -point swing. We report the conservative 50.45% profile and treat this variance as a finding (§8.1). For the record, an earlier project figure was generated from hand-coded placeholder values for models never trained here; it is not used—Figure 2 replaces it with real, provenance-marked numbers.

Model	Params	Family	Provenance
BERT-base	110M	encoder	M / M
DeBERTa-v3-large	435M	encoder	P / P
Qwen2.5-0.5B	0.5B	decoder	M / M
Qwen2.5-3B	3B	decoder	M / est.

Table 3: Models and per-task provenance (CQA / SIQA). M = measured, P = provisional, est. = estimated (§8.2). All fine-tuned runs: max length 128, peak LR 2×10^{-5} , linear warmup 10% then decay, weight decay 0.01, label smoothing 0.1, early stopping (patience 3), best checkpoint by validation accuracy. The 3B model uses Adafactor (Shazeer and Stern, 2018) (batch 1×8 accumulation) to fit one 12 GB GPU; smaller models use AdamW (Loshchilov and Hutter, 2019) (batch 8) with layer-wise LR decay.

5 Results

5.1 RQ1: scale compresses, not reshapes

On CQA, scaling the decoder 0.5B $\rightarrow$ 3B lifts overall accuracy from 68.63% to 84.93% (+16.3, two-proportion $z=9.5$, $p < 10^{-6}$; Table 4). The interesting result is its shape. The rank-ordering of category difficulty is near-perfectly preserved: the hardest categories for Qwen-0.5B (Spatial 59.1, Comparative 61.6, Negation 63.1) stay relatively hardest for Qwen-3B (Spatial 77.3, Negation 80.9, Comparative 82.2), and the easiest stay easiest. What changes is the floor: added scale is a broad, order-preserving lift that slightly flattens relative difficulty rather than eliminating any one failure mode. The category a small model finds hardest is still, proportionally, what the large model finds hardest.

The architecture comparison must respect the scale confound. BERT-base (110M) trails Qwen-0.5B (500M) everywhere ($z=9.2$, $p < 10^{-6}$ overall), but this conflates architecture with a $4.5\times$ parameter gap. The scale-controlled contrast is DeBERTa (435M) vs Qwen-0.5B (500M), where the provisional DeBERTa figures point to a large encoder advantage (84.0 vs 68.6 on CQA; 82.2 vs 72.0 on SIQA), concentrated in structured categories (negation, spatial). We flag this as the single result most worth re-running: if it holds, a disentangled-attention encoder is a more parameter-efficient commonsense reasoner than a decoder-classifier at matched scale. Notably, measured Qwen-3B (84.93) and provisional DeBERTa (84.03) are statistically indistinguishable overall ($z=0.61$, $p=0.54$), consistent with scale saturation—but this compares a measured to a provisional number. Following Dror et al. (2018), McNemar’s paired test is the appropriate choice once per-example predictions are regenerated.

Category	BERT [M]	Q-0.5B [M]	Q-3B [M]	DeB. [P]
Overall	50.45	68.63	84.93	84.03
Negation	45.22	63.06	80.89	85.35
Spatial	51.52	59.09	77.27	83.33
Causal	49.54	70.64	85.32	87.16
Temporal	45.26	66.42	84.67	84.67
Social	52.38	68.25	84.92	87.30
Comparative	45.21	61.64	82.19	82.19
Quantifier	52.49	64.64	81.22	80.11
Baseline	51.06	70.38	86.42	83.14

Table 4: CommonsenseQA category accuracy (%). [M] measured, [P] provisional (DeBERTa: reported, no artifact retained). Bold = best in row. Difficulty ordering is preserved across the two measured decoders.

Category	BERT [M]	Q-0.5B [M]	DeB. [P]
Overall	62.23	71.95	82.19
Negation	61.69	76.12	79.10
Spatial	61.06	70.65	81.56
Causal	63.78	73.14	82.86
Temporal	62.78	73.04	82.56
Social	62.64	72.92	82.63
Comparative	62.26	70.44	83.02
Quantifier	62.28	69.75	83.99
Baseline	58.82	65.20	81.86

Table 5: SocialIQA category accuracy (%). Qwen-3B on SIQA was not run; its estimate is in §8.2, kept out of this measured table.

5.2 RQ2: domain-transfer asymmetry

The small encoder is markedly better at social than physical commonsense: BERT-base scores 62.23% on SIQA vs 50.45% on CQA (+11.8), and Qwen-0.5B shows the same direction (71.95 vs 68.63; $z=6.5$, $p < 10^{-6}$). Two readings coexist and we cannot fully separate them: a genuine representational preference (social regularities may be more recoverable from pretraining than ConceptNet relations) and a task-structure effect (3-way vs 5-way chance rates differ by 13.3 points). Because the stronger decoder shows a much smaller gap (+3.3), we read the asymmetry as at least partly representational and concentrated in weaker models—i.e. the small-encoder profile is domain-sensitive, and that sensitivity shrinks with capability.

6 Error Analysis

The category view (Tables 4–5) exposes three patterns the overall number hides.

A spatial bottleneck scale does not close. Spatial is the lowest-scoring category on CommonsenseQA for both measured decoders: 59.09 for Qwen-0.5B (9.5 points below its overall 68.63) and 77.27 for Qwen-3B (7.7 below

its 84.93). A 16.3-point overall lift therefore recovers under two points of spatial’s relative deficit. Spatial reasoning in CQA requires resolving a frame of reference rather than retrieving a relation, and added parameters do not appear to supply that operation. Spatial is also the smallest CQA slice (5.4%, ≈ 66 instances), so we treat the ordering as robust and the exact margins as indicative.

Negation is task-dependent, not uniformly hard. On CommonsenseQA, Negation sits below overall accuracy for every model (BERT −5.2, Qwen-0.5B −5.6, Qwen-3B −4.0). On SocialIQA the sign flips for the decoder: Qwen-0.5B scores 76.12 on Negation against 71.95 overall (+4.2), its strongest category. The same regexes fire in both datasets, so this is a property of the register, not of the tagger—SocialIQA’s negation is largely conversational (“was not home”) rather than scope-bearing, and decoder pretraining appears to supply strong heuristics for it. Negation is a hard operation in CQA’s construction, not a hard operation in general.

The residual Baseline inverts across tasks. On CommonsenseQA the untagged Baseline sits above every model’s overall accuracy (51.06 / 70.38 / 86.42). On SocialIQA it is the hardest slice for both measured models and below overall accuracy for all three (58.82 / 65.20 / 81.86). The taxonomy is doing opposite work in the two benchmarks: in CQA the marked phenomena identify genuinely harder items, while in SocialIQA—where 89.6% of contexts fire at least one tag—the untagged residue is a small, atypical set (10.4%) of contexts stripped of the social and temporal cues the task is built around. A taxonomy calibrated on one benchmark’s marked phenomena should not be assumed to transfer its difficulty semantics to another.

7 Discussion

The single-number reading here is “bigger, better-architected models score higher”—true and uninformative. The category view converts that into structure: scale delivers a broad, order-preserving lift, not selective repair of specific bugs (§5.1).

To fix spatial reasoning, buying parameters is an inefficient lever: the category improves roughly in proportion to everything else and remains the floor at 3B. At this resolution we observe no selective repair anywhere in the profile—every category moves together—so we can neither confirm nor localize threshold effects of the kind the emergence debate concerns (Wei et al., 2022; Schaeffer et al., 2023). Detecting them would require finer slices than the per-category sample sizes here support, which is precisely the trade we have made. If the provisional encoder

Category-Level Accuracy Across Architecture and Scale

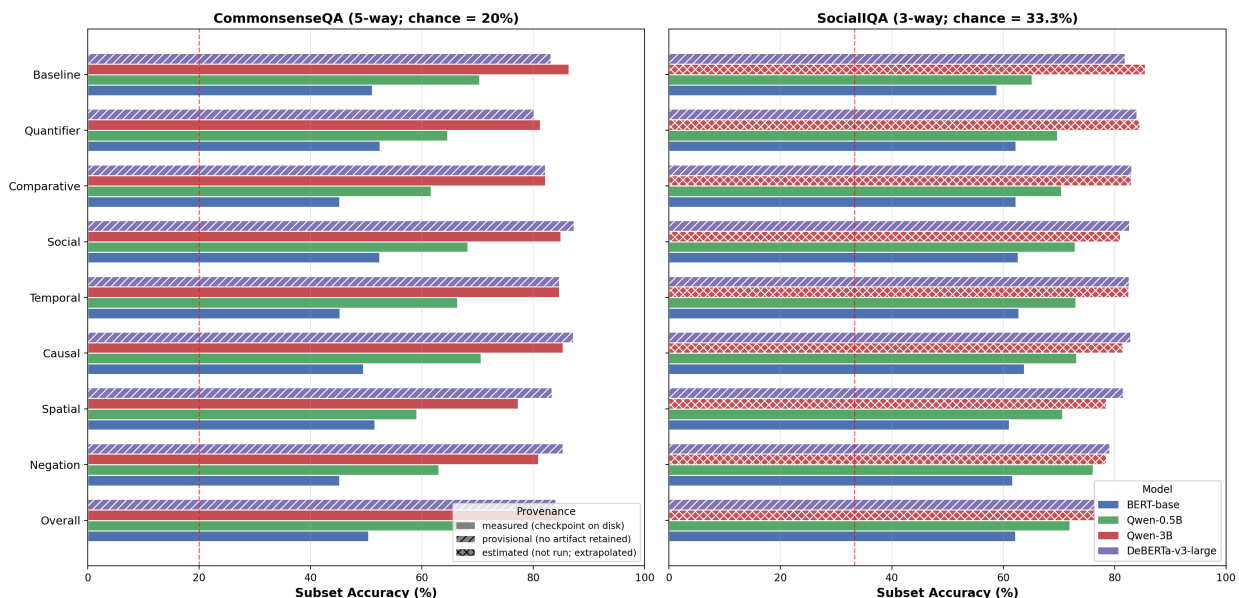


Figure 2: Category-level accuracy with provenance encoded in the bars: *solid* = measured, *hatched* = provisional (DeBERTa) or estimated (Qwen-3B/SIQA). Red dashed line = random-choice rate (20% CQA, 33.3% SIQA). The reader is never invited to treat an unverified cell as a measurement.

result survives re-running, the more actionable lever is *architectural*: a 435M encoder rivaling a 3B decoder, trained on one consumer GPU, is a strong parameter-efficiency statement—concentrated exactly where bidirectional context and relative-position modeling should help.

8 Limitations

(1) Provenance. DeBERTa (both tasks) and Qwen-3B/SIQA are not measured; dependent claims—especially encoder parameter-efficiency—are marked and must be re-run. **(2) Surface tagging.** The regex tagger matches keywords without word-sense disambiguation, so “*inside* his mind” is mis-tagged spatial; noise is auditable but present. **(3) Slice size.** Two CommonsenseQA categories rest on small pools—Spatial (5.4%, ≈ 66 instances) and Comparative (6.0%, ≈ 73)—so we treat single-category deltas below roughly five points as non-significant. We evaluate at parent level specifically to avoid the sub-20-instance cells a finer split produces on CQA. **(4) Confounds.** The 3B model uses a different optimizer, and encoder-vs-decoder entangles architecture with head design and (for DeBERTa) LoRA vs full fine-tuning. **(5) Significance.** We report a two-proportion approximation; McNemar’s paired test (Dror et al., 2018) is preferred once predictions are regenerated.

8.1 Reproducibility

The ~ 8 -point BERT/CQA swing between reruns (§4) is a warning, not a footnote: when cells rest on tens of examples, an aggregate swing from a missing seed and silent checkpoint overwriting can flip a category comparison’s sign. Trustworthy diagnostic evaluation requires seed control, checkpoint versioning, and per-cell variance. We treat explicit provenance tagging as part of the contribution because the diagnostic setting is unusually easy to over-read.

8.2 Estimating the missing cell

Rather than invent a measurement for the unrun Qwen-3B/SIQA cell, we report an explicit extrapolation, never mixed into measured tables. Using CQA—where all three models are available—we compute per-category the fraction ρ_i of the Qwen-0.5B $\rightarrow$ stronger-model gap that the 3B jump closes, then transfer ρ_i to the SIQA Qwen-0.5B baseline. This lands Qwen-3B at ~ 80 – 84% overall on SIQA, on par with provisional DeBERTa and consistent with the measured CQA saturation pattern—but it is a projection, not evidence, and the *Baseline* cell (largest extrapolation) is least trustworthy. It should be replaced by a real run before any load-bearing claim.

9 Conclusion

Aggregate accuracy is the wrong resolution for understanding commonsense reasoning. Our deterministic taxonomy turns a benchmark number into a diagnostic profile. On measured results, scaling a decoder 0.5B $\rightarrow$ 3B adds 16 points but preserves the ordering of category difficulty—scale compresses the profile rather than reshaping it—while spatial reasoning stays hardest at every scale, and a small encoder transfers better to social than physical commonsense. Provisional results suggest, but do not yet establish, that a mid-scale encoder rivals a $7\times$ larger decoder. Throughout, we treat provenance as first-class, because the same slicing that makes diagnostic evaluation informative also shrinks the sample behind every number.

References

- Yonatan Belinkov and James Glass. 2019. Analysis methods in neural language processing: A survey. *Transactions of the Association for Computational Linguistics (TACL)*, 7:49–72.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. The hitchhiker’s guide to testing statistical significance in natural language processing. In *Proceedings of ACL*, pages 1383–1392.
- Allyson Ettinger. 2020. What BERT is not: Lessons from a new suite of psycholinguistic diagnostics for language models. *Transactions of the Association for Computational Linguistics (TACL)*, 8:34–48.
- Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *International Conference on Learning Representations (ICLR)*.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2021. DeBERTa: Decoding-enhanced BERT with disentangled attention. In *International Conference on Learning Representations (ICLR)*.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, et al. 2022. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Edward J Hu, Yelong Shen, Phillip Wallis, et al. 2021. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*.
- Jared Kaplan, Sam McCandlish, Tom Henighan, et al. 2020. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*.
- Nora Kassner and Hinrich Schütze. 2020. Negated and misprimed probes for pretrained language models: Birds can talk, but cannot fly. In *Proceedings of ACL*, pages 7811–7818.
- Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*.
- Tom McCoy, Ellie Pavlick, and Tal Linzen. 2019. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of ACL*, pages 3428–3448.
- Qwen Team. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.
- Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of ACL*, pages 4902–4912.
- Maarten Sap, Hannah Rashkin, Derek Chen, Ronan Le Bras, and Yejin Choi. 2019. Social IQa: Commonsense reasoning about social interactions. In *Proceedings of EMNLP-IJCNLP*, pages 4463–4473.
- Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. 2023. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Noam Shazeer and Mitchell Stern. 2018. Adafactor: Adaptive learning rates with sublinear memory cost. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 4596–4604.
- Robyn Speer, Joshua Chin, and Catherine Havasi. 2017. ConceptNet 5.5: An open multilingual graph of general knowledge. In *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence*, pages 4444–4451.
- Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. 2019. CommonsenseQA: A question answering challenge targeting commonsense knowledge. In *Proceedings of NAACL-HLT*, pages 4149–4158.
- Jason Wei, Yi Tay, Rishi Bommasani, et al. 2022. Emergent abilities of large language models. *Transactions on Machine Learning Research (TMLR)*.
- Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. HellaSwag: Can a machine really finish your sentence? In *Proceedings of ACL*, pages 4791–4800.